HDBUDDYPRO

BY TEAM KAMPONG CREW

Lim Wei Sheng
Yun Kar Kit
Kee Yu Kai
Loh Han Meng
Tai Yi Sheng Winston
Lim Jun Jie

BACKGROUND

- Purchasing a home is a major milestone for many Singaporeans.
- HDB resale flat information is scattered across multiple government platforms.
- The volume of information makes it overwhelming and time-consuming to find, compare and shortlist suitable flats.

TARGET AUDIENCE

- First-time homebuyers
- Couples and families with children
- Budget-conscious buyers
- Existing homeowners looking to relocate

HDBUDDYPRO

A web platform that consolidates HDB resale flat data, nearby amenities and regional statistics, helping home seekers compare options based on their budget and lifestyle needs.

PROJECT OBJECTIVES

SEARCH AND FILTER

an intuitive filtering and map-based interface that lets users narrow down HDB flats by region, price, room type and remaining lease.

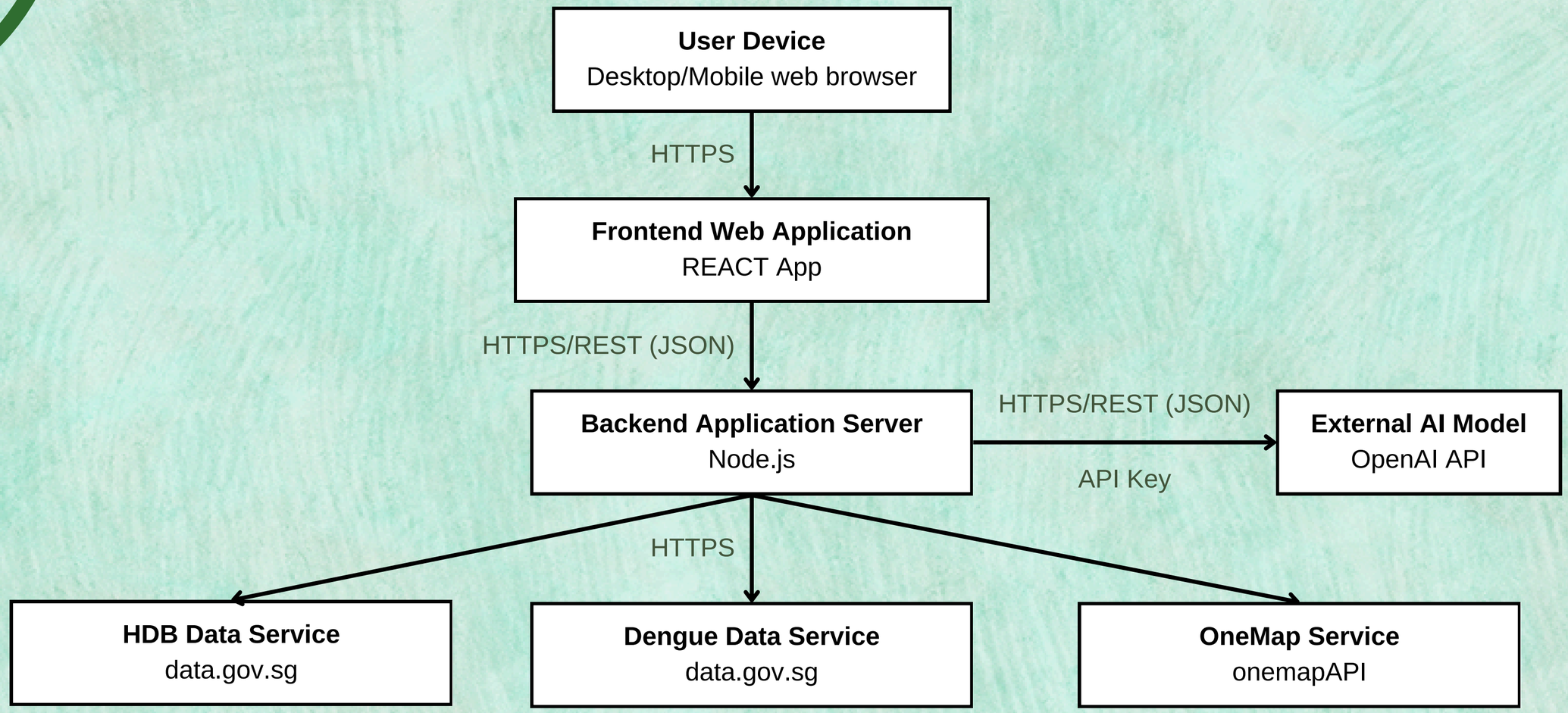
DATA INSIGHTS

Compare resale price distributions across regions using boxplots, and explore flat-type distributions through bar charts.

AI CHATBOT

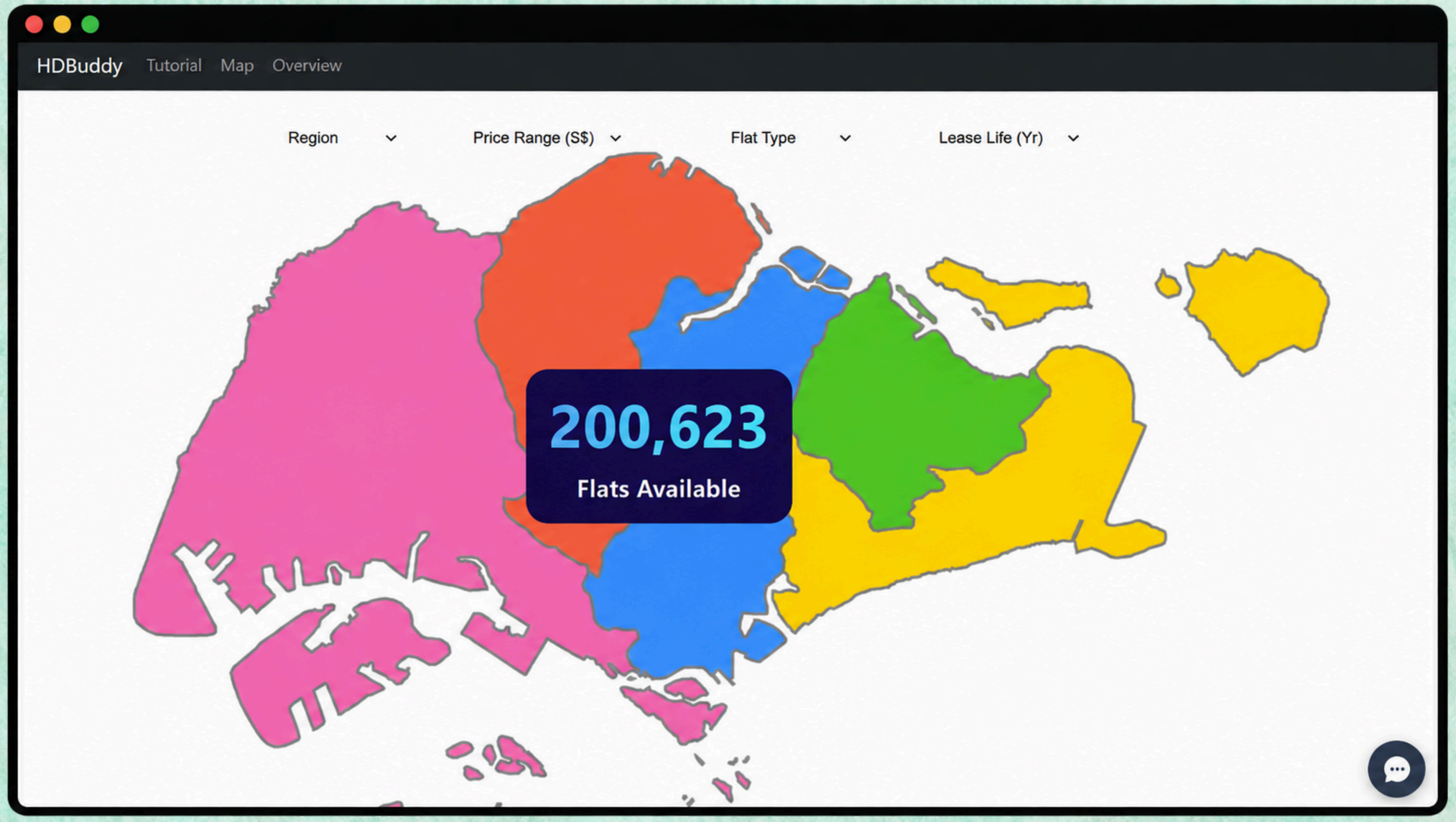
An AI-powered chatbot that allows users to ask questions about HDB flats and receive relevant responses based on available housing information.

SYSTEM ARCHITECTURE



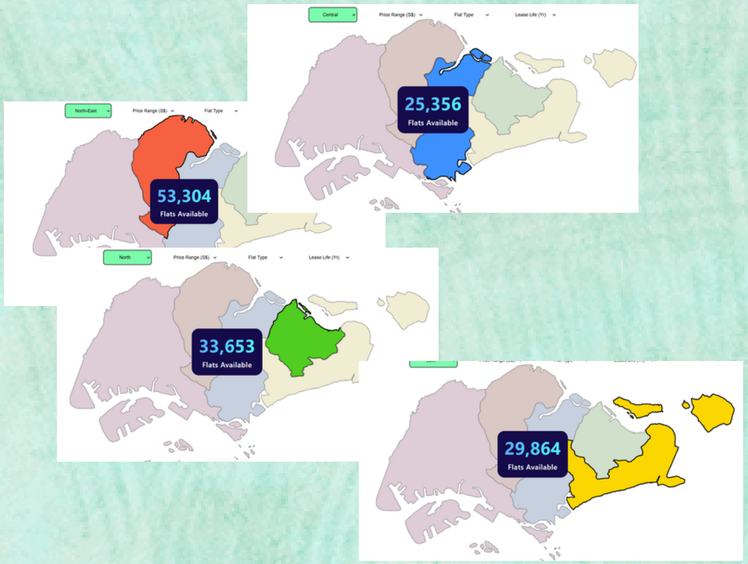
KEY FEATURES

HDBUDDYPRO MAP INTERFACE

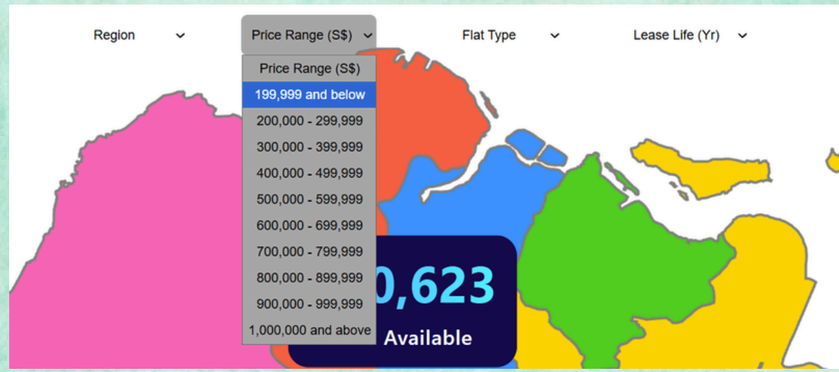


MAP FILTERING

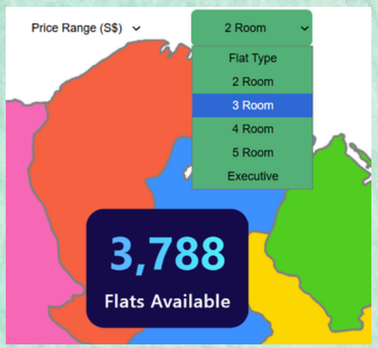
BY REGIONS



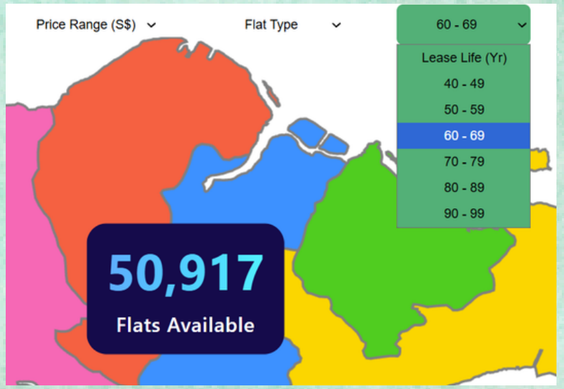
BY PRICE



BY FLATS

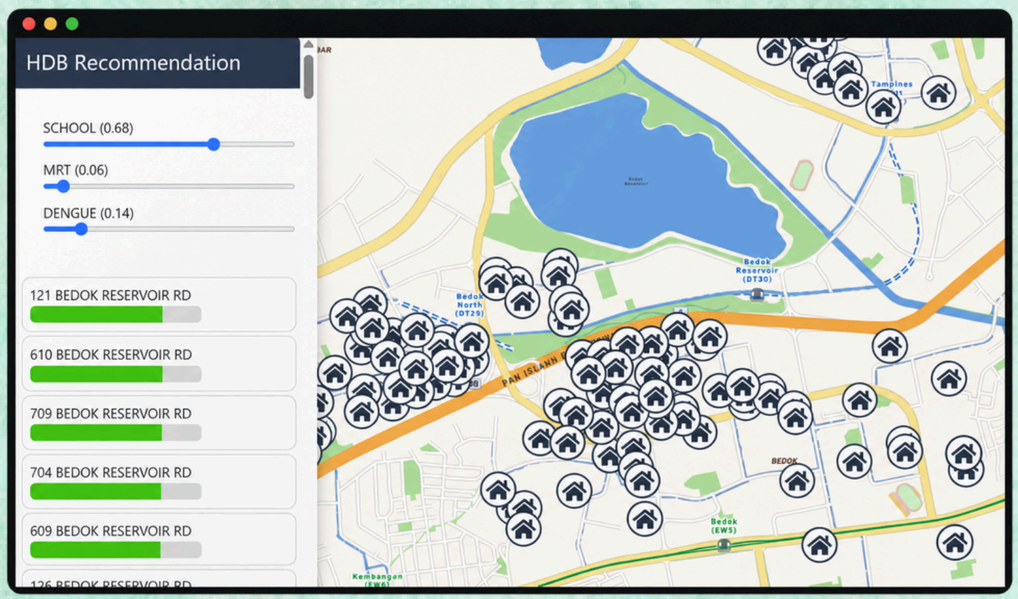


BY LEASE LIFE



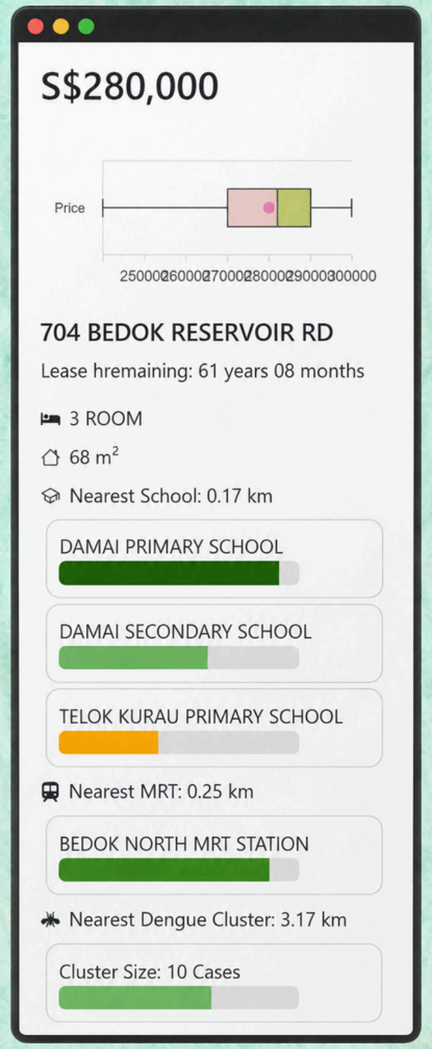
FINDING FLATS

HDB RECOMMENDATION



- On the homepage, users can narrow down HDB resale flats using filters such as region, price range, flat type, and remaining lease.
- After clicking **Find**, matching flats appear on the map.
- Users can adjust the **School, MRT, and Dengue** sliders to change which proximity factors matter most.
- The green progress bar shows a **proximity score.**

FLAT DETAILS

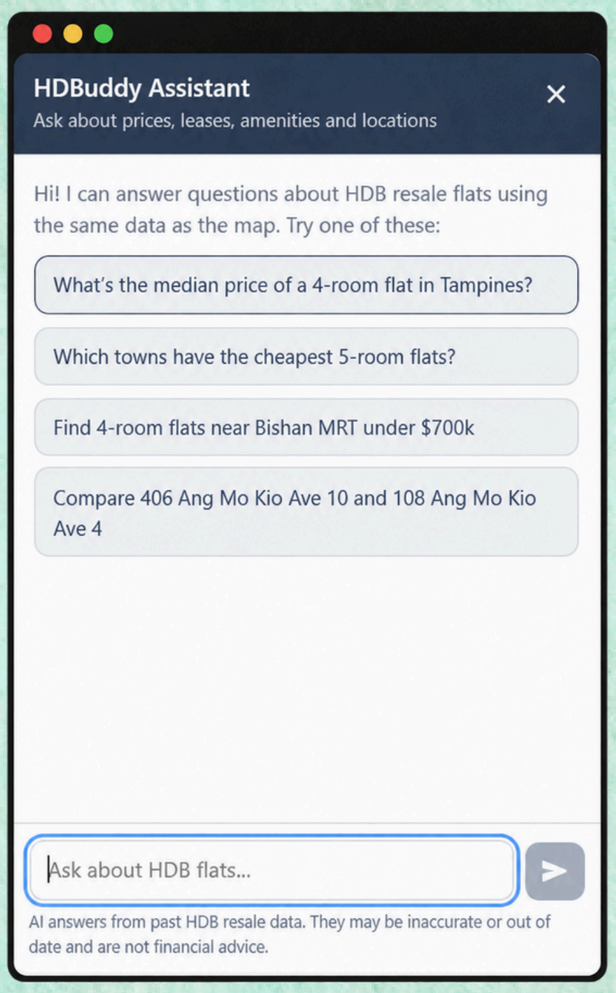


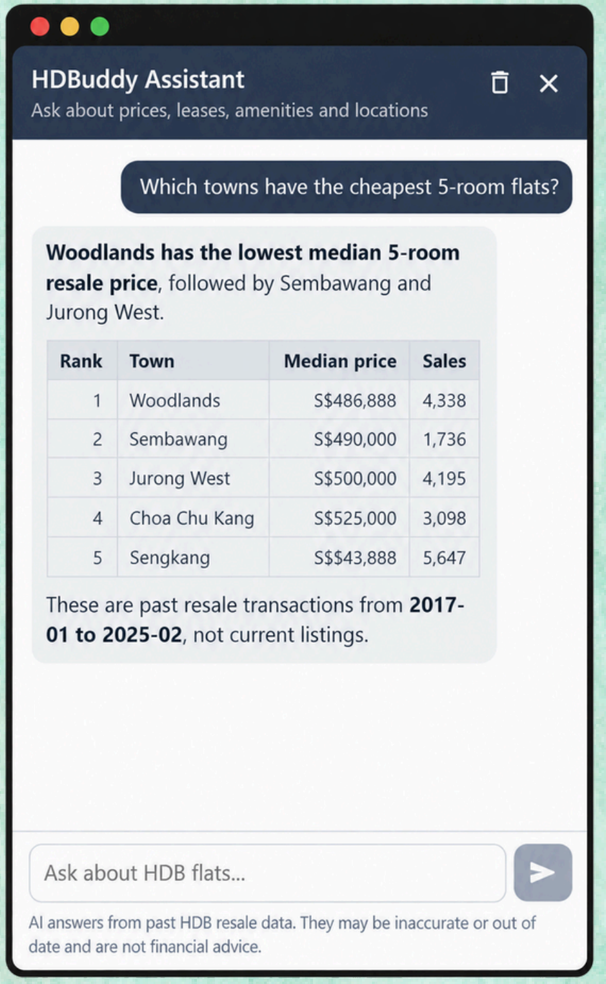
- Selecting a flat opens its information panel.
- Users can review the resale price, price distribution, remaining lease, flat type, and floor area.
- The panel also shows nearby schools, MRT stations, and dengue clusters, with estimated distances.
- This helps users compare a flat's price and location-related factors before deciding which options to explore further.

DATA ANALYTICS PRICE COMPARISON

As part of our data analytics, we use interactive box plots to compare HDB resale price distributions across regions and flat types, helping users understand median prices, price variations and areas that match their budget.

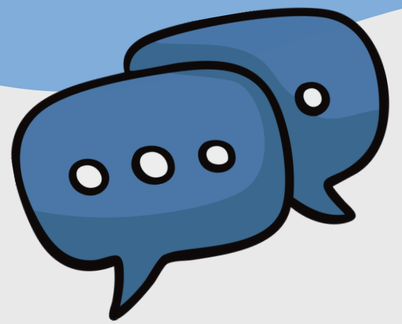
AI CHATBOT

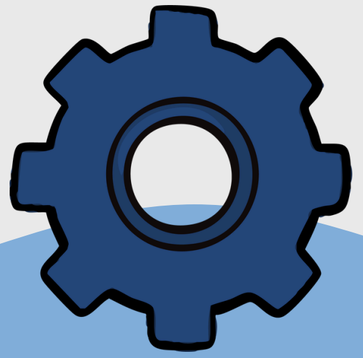




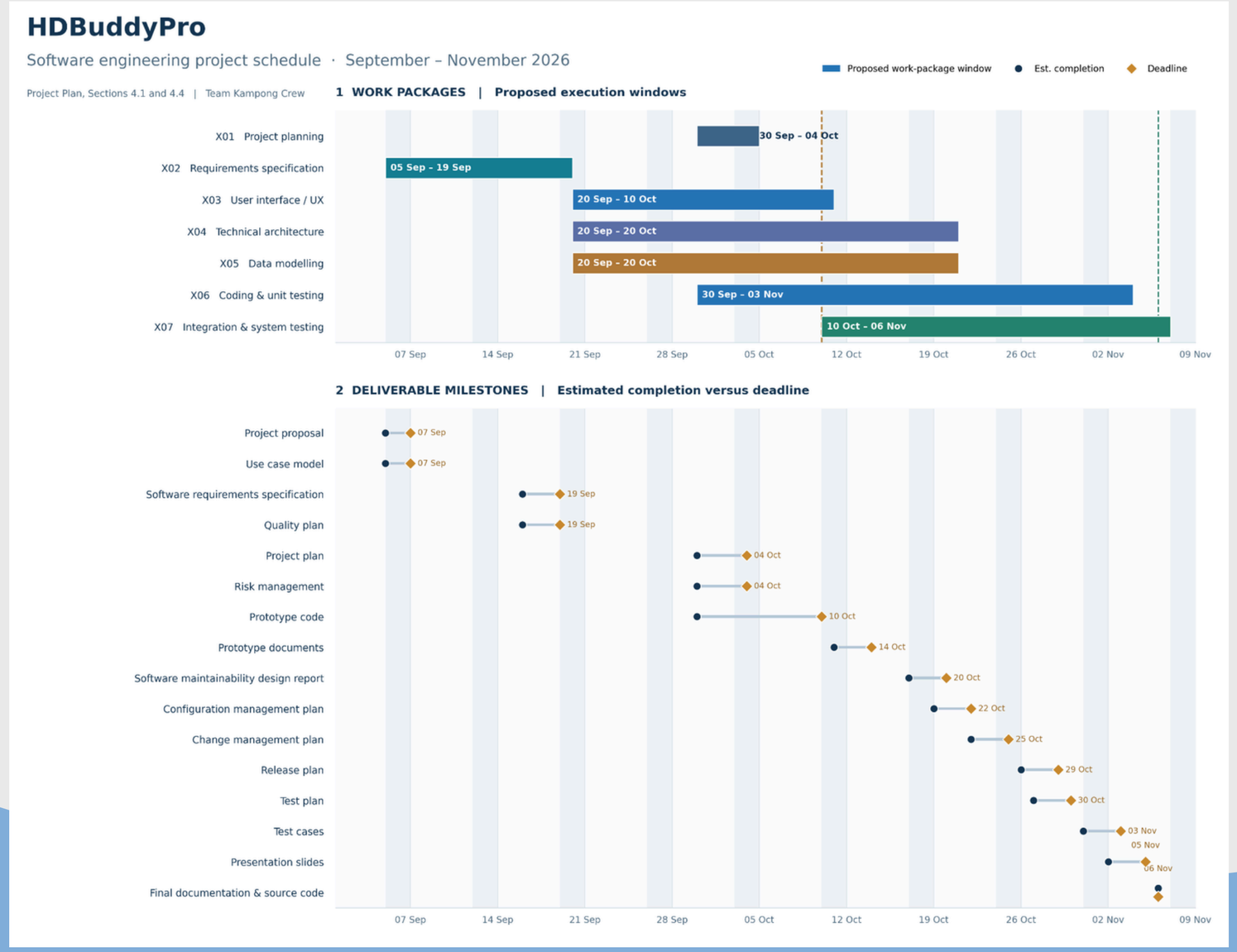
- Lets users ask natural-language questions about HDB resale flats
- Finds matching sales by criteria such as town, region, flat type, price, lease, and street.
- Summarizes resale prices and compares price statistics by town, region, flat type, or year.

Development Process





Project Schedule



Lab 1 - Project Planning and Scope Definition

1

Achievements

- Brainstormed HDBuddyPro and defined its objectives and core features.
- Completed the Project Proposal, Use Case Diagram and Use Case Descriptions.
- Created the team backlog and assigned responsibilities.

2

Issues

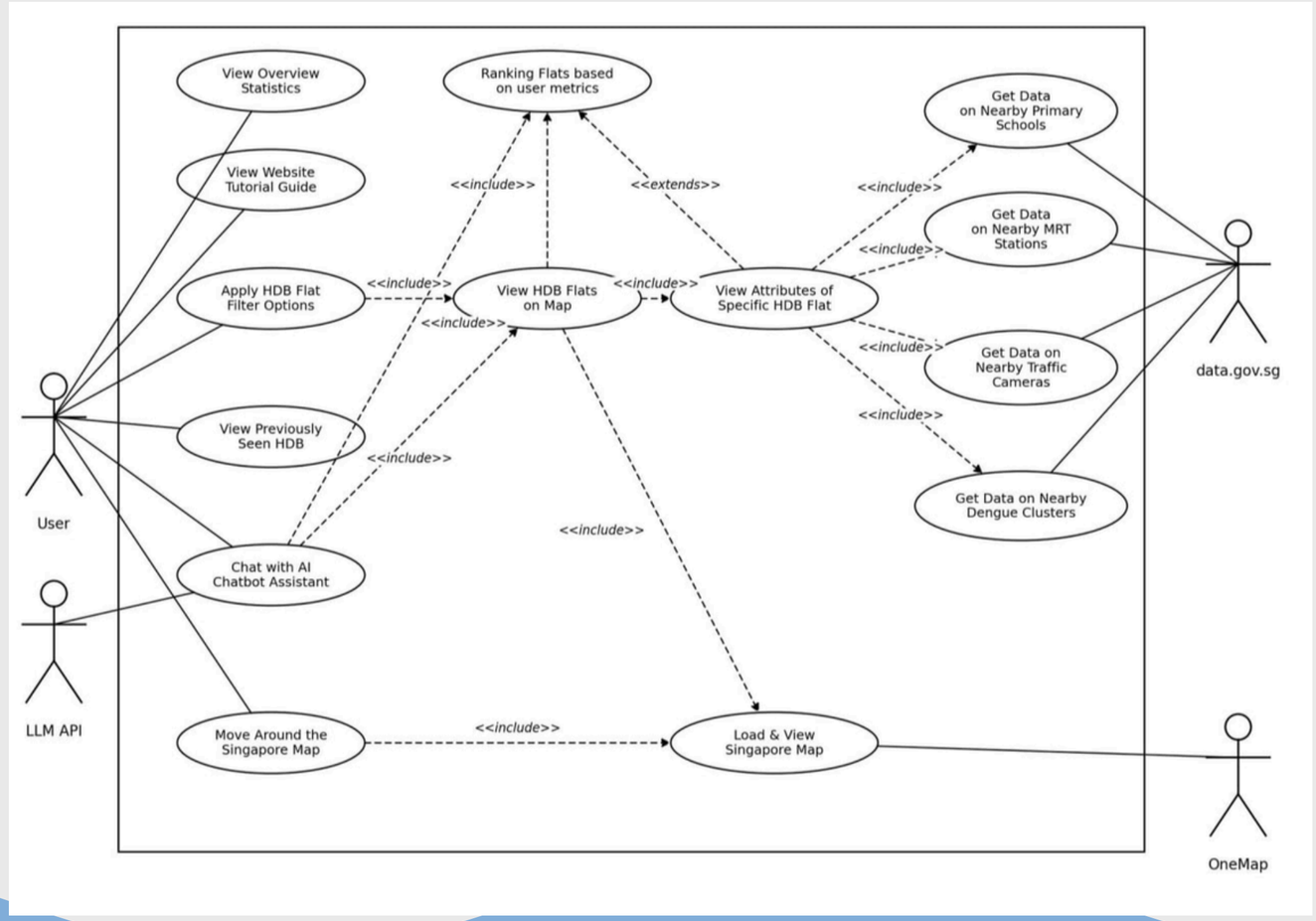
- Narrowing the project idea into a manageable scope.
- Different project ideas made it challenging to agree on a common direction.

3

Solutions

- Used the project proposal as a working draft to capture initial ideas, then refined and consolidated them through further team discussions to agree on the final scope and features.

Lab 1 - Project Planning and Scope Definition



Lab 2 - Requirements and Quality Planning

1

Achievements

- Developers discussed the technology stack and planned the initial prototype.
- Completed the System Requirements Specification and Quality Assurance Plan.
- Updated the team backlog with prototype development tasks.

2

Issues

- Need to turn feature ideas into clear requirements that developers could follow.
- Needed to define measurable quality standards for features such as the interactive map, personalised ranking and chatbot.

3

Solutions

- Discussed each feature's expected behaviour and documented it in the SRS to align the team's understanding.
- Defined component quality expectations and selected metrics such as test case pass rate, user satisfaction and response time

Lab 3 - Prototype Dev and Risk Management

1

Achievements

- Continued prototype development for frontend and backend, and reviewed progress across core features such as searching and filtering
- Completed the Project Plan, Risk Management Plan and Risk Log.

2

Issues

- Frontend and backend API communication was not working initially, preventing core features from retrieving and displaying data.
- Difficulty ranking risks that had the same likelihood and impact ratings.

3

Solutions

- Used AI to assist in debugging API errors for frontend-backend mismatches, then corrected the requests and retested data retrieval and display.
- Used the probability-impact matrix to assess priority, then proposed rankings based on effects on core features.

Lab 3 - Prototype Dev and Risk Management

Rank	RID	Risk Description	Probability	Impact	Risk Mitigation / Response
1	R01	**External API unavailability:** Data.gov.sg or OneMap services become unavailable, disrupting housing information retrieval or map functionality.	Medium	High	Implement error handling and use previously validated datasets where permitted.
2	R02	**Inaccurate or outdated housing information:** Government datasets contain incomplete, inconsistent or outdated data, affecting the accuracy of displayed flat information.	Medium	High	Validate retrieved data, monitor dataset updates and display data freshness information.
3	R03	**Frontend and backend integration failures:** Integration problems between React, Express.js, MongoDB and data services affect core application functionality.	Medium	High	Establish API contracts and conduct integration testing early.
4	R04	**Development schedule delays:** Underestimated effort or unforeseen technical difficulties affect completion of project milestones.	Medium	High	Track progress using GitHub Kanban, prioritise critical tasks and reallocate resources where necessary.
5	R05	**Inaccurate AI chatbot responses:** The chatbot generates incorrect or misleading housing information.	Medium	High	Validate responses against available housing data and restrict unsupported information.
6	R06	**Incorrect personalised ranking:** Errors in ranking calculations lead to results that do not accurately reflect user preferences.	Medium	Medium	Test ranking algorithms using predefined input weights and expected outputs.
7	R07	**Limited team availability:** Academic commitments or unexpected absences reduce available development resources.	Medium	Medium	Maintain shared documentation, communicate progress and redistribute critical tasks if necessary.
8	R08	**Insufficient testing time:** Development delays reduce the time available for integration, system and acceptance testing.	Medium	Medium	Begin testing early and prioritise test cases covering critical functionalities.
9	R09	**Application performance issues:** Large datasets or numerous map markers cause slow loading, filtering or rendering.	Medium	Medium	Optimise data queries, reduce unnecessary processing and test performance with realistic datasets.
10	R10	**Unexpected AI API or hosting costs:** Actual usage exceeds estimated resource requirements.	Low	Medium	Monitor API consumption and expenese, apply usage limits and review hosting requirements.

Lab 4 - Integration and Demo Preparation

1

Achievements

- Completed HDBuddyPro's development and integrated all core features.
- Completed the Release Management Plan, Cofiguration and Change Management Plan.
- Conducted QA testing before project demo to ensure everything works well.

2

Issues

- Some features did not behave as expected during QA testing, requiring fixes before the demonstration.

3

Solutions

- Defined QA checks before release, prioritised critical fixes, and planned to exclude unresolved features while documenting their limitations.

LEARNINGS

- Applying advanced software engineering and project management practices to develop HDBuddyPro.
- Understanding the entire process of how requirements, quality assurance, risk management and change control guide development and release decisions.
- Recognised the importance of collaboration between different stakeholders and how it can affect realistic project planning, clear responsibilities and time for testing.

Q&A